NAME: _____ Geometry (FL B.E.S.T.)

2.3 Theorems about Perpendicular Lines

Perpendicularity is a stronger statement than just "one right angle" — it forces all four angles at the intersection to be $90°$, which is what lets a single right-angle mark unlock an entire figure. This section links perpendicularity to parallelism in two directions: two lines each perpendicular to a third must be parallel to each other, and a line perpendicular to one of two parallel lines is automatically perpendicular to the other (the Perpendicular Transversal Theorem). Expect a short proof, not just a fact to quote, along with algebra where the right-angle condition itself becomes an equation.

Practice Set — All Questions

15 questions. Work them on your own paper, then check the key below.

1. $\ell \perp m$ means they form what measure?

2. Perpendicular lines form how many right angles?

3. $a \perp c$ and $b \perp c$. Then a and b are?

4. $\ell_1 \parallel \ell_2$ and $t \perp \ell_1$. Then t and ℓ_2?

5. Perpendicular intersection; one angle is?

6. Right angle's linear-pair partner?

7. Horizontal line and vertical line are always?

8. Symbol for perpendicular?

9. Symbol for parallel?

10. Two lines each $\perp$ to same line. They are?

11. Perpendicular lines; one angle measures x. Find x.

12. Parallel lines; transversal $\perp$ one. It is $\perp$ the other?

13. Can two distinct lines both $\perp$ the same line and also intersect each other?

NAME: _____ Geometry (FL B.E.S.T.)

14. Lines a and b are both perpendicular to line c. Explain in one sentence, using corresponding angles, why a must be parallel to b.

15. Transversal t meets ℓ at an angle marked $(4x - 6)^\circ$ as a right angle, and $\ell \parallel m$. Find x, and state the measure of the angle t forms with m.

✓ Answer Key

1. 90°
2. 4
3. Parallel
4. Perpendicular
5. 90°
6. 90°
7. Perpendicular
8. $\perp$
9. $\parallel$
10. Parallel
11. $x = 90$
12. Yes
13. No
14. Both a and b meet c at 90° in matching corresponding positions, so those corresponding angles are congruent, and by the Converse of the Corresponding Angles Postulate $a \parallel b$
15. $x = 24$; t meets m at 90°